

Detailed testing of the Western Leopard Toad (*Sclerophrys pantherina*)

The Western Leopard Toad is a narrow endemic restricted to a small region of the Western Cape, South Africa, with a total range of ~170 km. Within this range, it occurs in two disjunct populations about 100 km apart: the Cape Peninsula and the Overstrand (Measey & Tolley, 2011). The species is listed as Endangered on the IUCN Red List due to multiple threats. The Cape Peninsula population is heavily impacted by urbanisation, including roads, habitat fragmentation, pollution, and invasive species, which limit breeding and dispersal (Measey et al., 2019; Rebelo et al., 2011), although artificial water bodies may facilitate some connectivity (Measey & Tolley, 2011). The Overstrand region is less urbanised but has experienced substantial agricultural and residential transformation, leading to habitat loss and the decline of historical breeding sites; however, dispersal potential may still support gene flow among sites (Martha Smith & David Green 2005; Raymond Semlitsch 2008; Ingrid Van Bocxlaer et al. 2010; John Measey & Krystal Tolley 2011).

Please refer to a detailed description of evidence used for the Western Leopard Toad to DOI: [insert link DOI](#).

1. Fill in species and scenario information.

We tested four three candidate scenarios : (1) a single metapopulation (null hypothesis), (2) two subpopulations representing the Cape Peninsula and the Overberg region, and (3) three subpopulations, where the northern locality of this species on the Cape Peninsula is further separated from the remaining southern distribution.

Phase 1: Identify Subpopulations

This survey is best views on Desktop, or rotate your phone for a better view.

Please use the dropdown menu below for additional information.

- Subpopulations
- Evolutionarily Significant Units (ESU)
- Possible Evolutionarily Significant Unit (pESU)
- Gene Flow
- Potential IUCN Species assessment
- References

Scoring:

- 2 points or higher = distinct subpopulations
- Less than 2 points = stop assessment

Lines of evidence

Which species are you testing? *

Sclerophrys pantherina

How to set up scenarios to test:

This survey allows the evaluation of **multiple subpopulation scenarios**.

The **first scenario always represents the null hypothesis**, which assumes a **single metapopulation**. This one-metapopulation model must be tested first and serves as the baseline against which all alternative scenarios involving multiple subpopulations are compared.

When selecting scenarios that include **multiple subpopulations** (i.e. not the default metapopulation), **all subpopulations must be mutually distinct and supported by evidence**. Each subpopulation must have a clear evidentiary basis for inclusion in the scenario being tested. If any two subpopulations cannot be distinguished from one another, they should be merged, and a revised scenario should be considered for testing.

How many Scenarios do you want to assess?

3 Scenarios

1 Metapopulation	2 Subpopulations	3 Subpopulations
Scenario 1 name	Scenario 2 name	Scenario 3 name

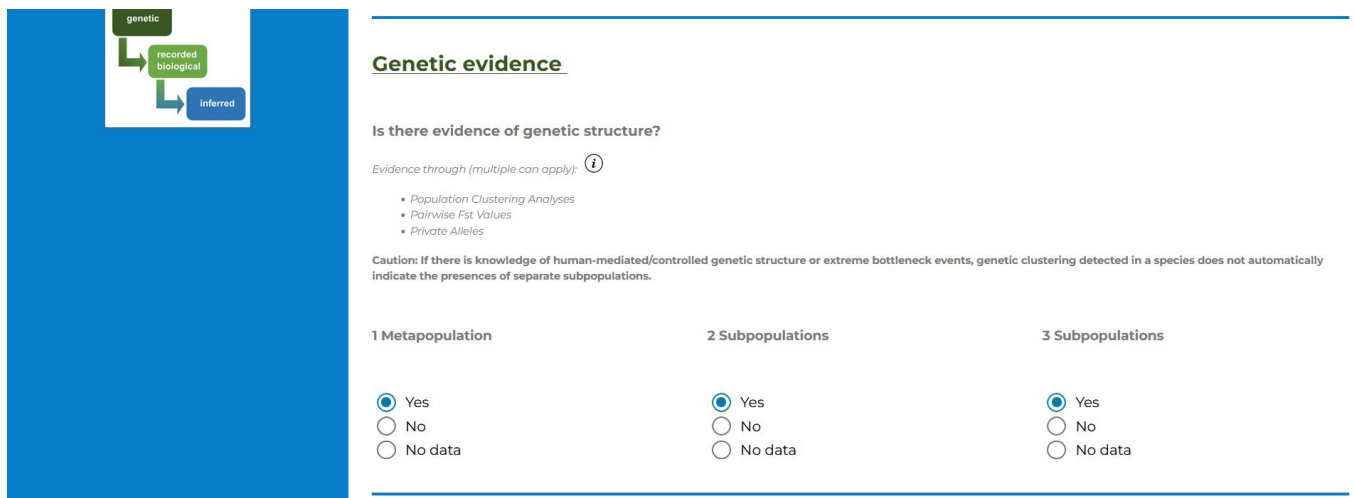
2. Answer the questions for each scenario.

a. Genetic evidence

In **Scenario (1)** (null hypothesis), depicting a single metapopulation, scoring is reserved to keep the questions identical across all scenarios. Mitochondrial DNA and microsatellite data did not support the presence of a single metapopulation, as strong genetic isolation was observed between the Cape Peninsula and the Overberg region with no shared haplotypes (Measey & Tolley, 2011; da Silva, unpublished data). Therefore, a score of 0/2 for genetic structure was assigned.

In **Scenario (2)**, genetic clustering analyses supported two subpopulations ($K = 2$), separating the Cape Peninsula from the Overberg region (Measey & Tolley, 2011). This resulted in a score of 2/2 for genetic structure.

In **Scenario (3)**, microsatellite data indicated a more complex structure with $K = 3$, including differentiation of the northern-most Cape Peninsula locality (Observatory) from both the Overberg and southern suburban localities (da Silva, unpublished data). Despite this finer-scale subdivision, clear genetic structure was still present, resulting in a score of 2/2 for genetic structure.



Genetic evidence

Is there evidence of genetic structure?

Evidence through (multiple can apply): 

- Population Clustering Analyses
- Pairwise F_{ST} Values
- Private Alleles

Caution: If there is knowledge of human-mediated/controlled genetic structure or extreme bottleneck events, genetic clustering detected in a species does not automatically indicate the presences of separate subpopulations.

1 Metapopulation	2 Subpopulations	3 Subpopulations
☒ Yes	☒ Yes	☒ Yes
☐ No	☐ No	☐ No
☐ No data	☐ No data	☐ No data

b. Recorded biological evidence

In **Scenario (1)** (null hypothesis), depicting a single metapopulation, scoring is reserved to keep the questions identical across all scenarios. The existing evidence did not indicate a single metapopulation. Therefore, a score of 0/2 for isolation/migration barriers was assigned.

In **Scenario (2)**, approximate Bayesian inference and spatial analysis of molecular variance indicate historical bottleneck events associated with sea level changes during interglacial periods, which likely isolated the Cape Peninsula and Overberg subpopulations (Measey & Tolley, 2011; Stephens et al., 2022). In addition, the historical loss of breeding sites further

supports long-term isolation (Measey & Tolley, 2011). Therefore, a score of 2/2 for isolation/migration barriers was assigned.

In **Scenario (3)**, although genetic structure exists between the Observatory locality and southern suburbs on the Cape Peninsula, this is interpreted as human-mediated due to the presence of major highways which began construction in the 1950s, which restrict gene flow for approximately 30+ generations. Therefore, this scenario does not reflect natural isolation processes and received a score of 0/2 for isolation/migration barriers.

Recorded biological evidence

Is there evidence of natural isolation or subpopulation fragmentation (through migration barriers)?

Evidence through (multiple can apply): i

- Occupation of different biogeographical zones
- Occupation of discrete remnants of historical habitats with little/no natural migration

Caution: In this framework, the natural fragmentation of subpopulations should be distinguished from human-mediated/controlled fragmentation of subpopulations, since the latter does not automatically indicate the presences of separate subpopulations.

1 Metapopulation	2 Subpopulations	3 Subpopulations
☒ Yes ☐ No ☐ No data	☒ Yes ☐ No ☐ No data	☐ Yes ☒ No ☐ No data

c. Inferred evidence

In **Scenario (1)** (null hypothesis), depicting a single metapopulation, scoring is reserved to keep the questions identical across all scenarios. Species distribution models across the range of WLT indicate a clear climatic and habitat suitability disjunction between the Cape Peninsula and Overberg subpopulations (Casola, 2017). This existing evidence did not indicate a single metapopulation. Therefore, a score of 0/2 for inferred evidence was assigned.

In **Scenario (2)**, the same modelled disjunction between the Cape Peninsula and Overberg subpopulations supports inferred geographic separation (Casola, 2017) and indicates two distinct subpopulations. This resulted in a score of 2/2 for the inferred evidence.

In **Scenario (3)**, no gaps in climatic suitability were detected within the Cape Peninsula between the southern suburbs and the northern Observatory site, indicating no inferred barrier at this finer scale. Therefore, this scenario received a score of 0/2 for inferred evidence was assigned.

Inferred evidence

Is there inferred evidence of likely subpopulation fragmentation?

Note: Only relevant if genetic or recorded evidence from the focal species is absent. Evidence of inferred patterns can be used. If nothing is known about the focal species, co-occurring or closely related species could give indication of subpopulation structure.

Evidence through (multiple can apply): ⓘ

- **inferred** subpopulation disjunction (e.g., species distribution modeling)
- Phylogeographic and/or biogeographic evidence **from co-occurring species**
- Assumption of biogeographical data/refugia **without direct evidence**
- **inferred** dispersal distance and buffer according to the CBD indicators

1 Metapopulation

- ☒ Yes
☐ No
☐ No data

2 Subpopulations

- ☒ Yes
☐ No
☐ No data

3 Subpopulations

- ☐ Yes
☒ No
☐ No data

3. Provide name and e-mail address (optional). This will allow the results to be sent to your email address.

Notes

Name & Surname

Jane Doe

Email Address

jane.doe@xxxx

Data protection disclaimer

By providing your name and email, you will receive the results in your inbox, as well as on the next page and may receive communications from us to participate further in the development of this framework. If you **choose not to enter your email**, your results will still be presented to you on the next page.

Submit Data

4. Click the **“Submit Data”** button to receive the results (mandatory to get results).

5. Interpreting the results.

This is the results page. It lists each question and the responses entered by the user as verification. Below the questions is a summary of the scores for each scenario.

For the Western Leopard Toad, two of the three candidate scenarios (**Scenarios (2) and (3)**) meet the minimum threshold of two points, and therefore qualify as subpopulation scenarios. The final tallies for Phase 1 however indicate that **Scenario (2)** (two subpopulations) received the highest score, suggesting that the presence of two naturally fragmented subpopulations is best supported. From a management perspective, **Scenario (3)** should still be considered, as

the Observatory population harbours unique genetic diversity that warrants protection. Given that re-establishing natural movement corridors between Observatory and the southern suburbs is unlikely, management efforts should focus on maintaining this isolated population. Alternatively, as all three subpopulations appear to be severely genetically compromised (Stephens et al., 2022; da Silva, unpublished data), genetic rescue through the managed exchange of genetic material within the Peninsula may be warranted.

Phase 1: Identify Subpopulations

This survey is best views on Desktop, or rotate your phone for a better view.

Please use the dropdown menu below for additional information.

Subpopulations

Evolutionarily Significant Units (ESU)

Possible Evolutionarily Significant Unit (pESU)

Gene Flow

Potential IUCN Species assessment

References

Scoring:

2 points or higher = distinct subpopulations

Less than 2 points = stop assessment

Lines of evidence

genetic

recorded

biological

Thank you , for participating in the CGSG Subpopulation and ESU survey.

The species you tested is the *Sclerophrys pantherina*

Your Survey Results:

1. Is there evidence of population structure? (2 points)

1 Metapopulation Yes	2 Subpopulations Yes	3 Subpopulations Yes	
-------------------------	-------------------------	-------------------------	--

2. Is there evidence of natural isolation or subpopulation fragmentation (through migration barriers)? (2 points)

1 Metapopulation Yes	2 Subpopulations Yes	3 Subpopulations No	
-------------------------	-------------------------	------------------------	--

3. Is there inferred evidence of likely subpopulation fragmentation? (1 points)

1 Metapopulation Yes	2 Subpopulations Yes	3 Subpopulations No	
-------------------------	-------------------------	------------------------	--

Total score for the scenario: 1 Metapopulation is 0

Total score for the scenario: 2 Subpopulations is 5

Total score for the scenario: 3 Subpopulations is 2

Total score for the scenario: is 0

For scenarios that scored 2 points or higher, please use the button below to continue to Phase 2.

GO TO PHASE 2>>

You'll be receiving the results in your inbox soon!

- Click on the “Phase 2” button to proceed.
- As in Phase 1, enter the species and scenario information. This does not get carried over from Phase 1. Based on the results of Phase 1, we tested three candidate scenarios for delineating Evolutionarily Significant Units (ESUs): (1) a single ESU, because subpopulations can be nested within an ESU (2) two ESUs (Cape Peninsula and Overberg), (3) three ESUs (Observatory, southern suburbs of Cape Town and Overberg).

Phase 2: Delineate Evolutionarily Significant Units (ESUs)

Please use the information provided by the email sent, to complete the following ESU survey.

Please use the dropdown menu below for additional information.

Subpopulations

Evolutionarily Significant Units (ESU)

Probable Evolutionarily Significant Unit (pESU)

Gene Flow

Potential IUCN Species assessment

References

Scoring:

10 points or higher = distinct ESU

Below 10 points:

a) 10 points cannot be reached or exceeded (even if all missing data were supportive of the tested scenario) =no support for the ESU - Scenario.

Which species are you testing?

Sclerophrys pantherina

How to set up scenarios to test:

This survey allows the evaluation of multiple ESU scenarios.

The first scenario always represents the null hypothesis, which assumes a single Evolutionarily Significant Unit (ESU). This one-ESU model must be tested first and serves as the baseline against which all alternative scenarios involving multiple ESUs are compared.

When selecting scenarios that include multiple ESUs (i.e. not the default single ESU), all ESUs must be mutually distinct and supported by evidence. Each ESU must have a clear evidentiary basis for inclusion in the scenario being tested. If any two ESUs cannot be distinguished from one another, they should be merged, and a revised scenario should be considered for testing.

How many scenarios are you testing?

3 Scenarios

1 ESU

2 ESUs

3 ESUs

Scenario 1 name

Scenario 2 name

Scenario 3 name

8. Answer the questions for each line of evidence for each scenario being investigated.

a. Genetic evidence

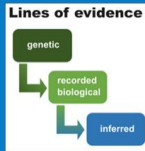
In **Scenario (1)** (null hypothesis), depicting a single ESU, scoring was reserved to maintain consistency across scenarios. No chromosomal or karyotypic information is available for this species, resulting in 0/6 for karyotype variation. Mitochondrial DNA and population-level analyses indicate strong genetic isolation between the Cape Peninsula and Overberg regions, with no shared haplotypes (Measey & Tolley, 2011). ABC and SAMOVA analyses have indicated that several bottleneck events associated with sea level rise during interglacial periods, contributed to the isolation of the Cape Peninsula and Overberg distributions (Measey & Tolley, 2011; Stephens et al., 2022), contradicting a single ESU and resulting in 0/6 for evolutionary distinctiveness. No evidence for adaptive divergence data is available for this scenario (0/6 points).

In **Scenario (2)**, representing two ESUs, chromosomal data remain unavailable (0/6 points). Genetic evidence still shows strong isolation between the Cape Peninsula and Overberg regions, supported by mitochondrial DNA and bottleneck analyses (Measey & Tolley, 2011; Stephens et al., 2022), resulting in 6/6 for evolutionary distinctiveness. No evidence for adaptive divergence data is available for this scenario (0/6 points).

In **Scenario (3)**, representing a finer subdivision including the Observatory population and southern suburbs, no chromosomal data are available (0/6 points). Genetic analyses indicate that the isolation of the Observatory population is recent and not supported by reciprocal monophyly relative to the southern suburb group (Measey & Tolley, 2011). This suggests incomplete lineage sorting rather than long-term evolutionary separation; therefore 0/6 points was assigned for evolutionary distinctiveness. No adaptive divergence data are available for this scenario (0/6 points).

by no points are currently not reached or exceeded, but could if missing data subsequently prove to be supportive of the tested scenario = scenario indicates possible ESUs (pESUs)*

*However, if a candidate scenario currently only scores in recorded biological evidence-types (so non-genetic), the scenario should be regarded as 'data-deficient' rather than as a pESU scenario.



Genetic evidence

Is there inherited variation in chromosome numbers, ploidy level or chromosome structure?

Evidence through (multiple can apply):

- Karyotype variation
- Difference in ploidy levels between units
- Structural variation of chromosomes between units

1 ESU

- ☐ Yes
☐ No
☒ No data

2 ESUs

- ☐ Yes
☐ No
☒ No data

3 ESUs

- ☐ Yes
☐ No
☒ No data

Is there evidence of long-term reproductive isolation?

Evidence through (multiple can apply):

- Reciprocal Monophyly
- Molecular Estimate of Divergence Time

1 ESU

- ☒ Yes
☐ No
☐ No data

2 ESUs

- ☒ Yes
☐ No
☐ No data

3 ESUs

- ☐ Yes
☒ No
☐ No data

Is there evidence of local adaptation or adaptive divergence?

Evidence through (multiple can apply):

- Genomic signals of local adaptation
- Evidence from hybrid zones

1 ESU

- ☐ Yes
☐ No
☒ No data

2 ESUs

- ☐ Yes
☐ No
☒ No data

3 ESUs

- ☐ Yes
☐ No
☒ No data

b) Recorded biological evidence

In **Scenario (1)** (null hypothesis), depicting a single ESU, scoring was reserved to maintain consistency across scenarios. WLTs exhibit strong breeding site fidelity, with adults returning annually to the same breeding wetlands and ponds (late July to September). This inherited behavioural trait may reinforce population structuring and limit dispersal and is contradicting a single ESU and resulting in 0/4 for inherited characteristic variation. No evidence was found for recorded variation or traditional knowledge in either scenario (0/2 each).

In **Scenario (2)**, strong breeding site fidelity is observed across the species, with adults returning to the same water bodies each breeding season, potentially reinforcing population structure. However, no evidence links this behaviour to distinct ESU-level differences. This information was nevertheless considered because breeding site fidelity is an inherited behavioural trait with the potential to influence genetic differentiation and local adaptation.

Since the behaviour appears consistent across the species rather than varying among ESUs, it did not provide support for distinguishing separate units and was therefore scored as 0/4 for inherited characteristic variation. No evidence was found for recorded variation or traditional knowledge in either scenario (0/2 each).

In **Scenario (3)**, the same strong breeding site fidelity is present across populations, but there is no evidence that this behaviour differs among finer-scale genetic subdivisions. This information was nevertheless considered because site fidelity can contribute to restricted dispersal and fine-scale structuring, even when not directly linked to adaptive differences. However, as the behaviour was shared broadly and did not distinguish among the proposed subpopulations, it did not support subdivision at this level and was therefore scored as 0/4 for inherited characteristic variation. No evidence was found for recorded variation or traditional knowledge in either scenario (0/2 each).

Recorded biological evidence

Is there evidence of inherited characteristic differences?

Evidence through (multiple can apply): 

- Heritable differences in focal traits observed in experiments or confirmed by analytical tests

1 ESU

- ☒ Yes
☐ No
☐ No data

2 ESUs

- ☒ Yes
☐ No
☐ No data

3 ESUs

- ☒ Yes
☐ No
☐ No data

Is there evidence supporting transmitted characteristic differences?

Evidence through (multiple can apply): 

- Cultural or learnt behavioral differences unique or specific between units
- Acquired or transmitted traits (e.g., migration pattern, methylation differences, body size differences)

1 ESU

- ☐ Yes
☐ No
☒ No data

2 ESUs

- ☐ Yes
☐ No
☒ No data

3 ESUs

- ☐ Yes
☐ No
☒ No data

Is there traditional knowledge suggesting distinctiveness between units?

Evidence through (multiple can apply): 

- Recorded differences based on traditional and local knowledge

1 ESU

- ☐ Yes
☐ No
☒ No data

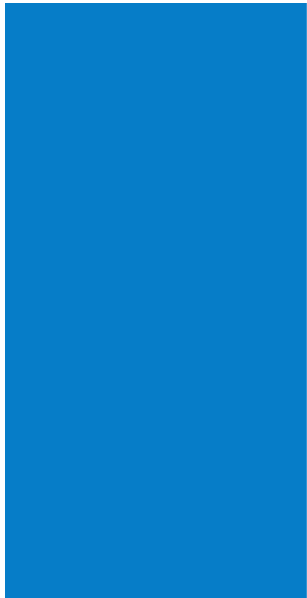
2 ESUs

- ☐ Yes
☐ No
☒ No data

3 ESUs

- ☐ Yes
☐ No
☒ No data

9. Enter your name and e-mail address (optional).



Notes

Name & Surname
Jane Doe

Email Address
jane.doe@xxxxx

Data protection disclaimer
By providing your name and email, you will receive the results in your inbox, as well as on the next page and may receive communications from us to participate further in the development of this framework. If you **choose not to enter your email**, your results will still be presented to you on the next page.

Submit Data

10. Click the “**Submit Data**” button to receive the results.

11. Interpreting the results.

As with the subpopulation survey results, each question and the answers provided are presented on the results page. A summary of the total scores for each scenario is presented below that.

The results of Phase 2 clearly identify two ESUs within the Western Leopard Toad (**Scenario (2)**; 10 points). The other scenarios were not supported, with **Scenario (1)** (a single ESU) receiving a score of zero and **Scenario (3)** a score of four points. As these scores are below the threshold of 10 and include ‘No data’ responses, **Scenario (3)** remains plausible (i.e., 3 pESUs). Should additional data become available, **Scenario (3)** should be re-examined.

Phase 2: Delineate Evolutionarily Significant Units (ESUs)

Please use the information provided by the email sent, to complete the following ESU survey.

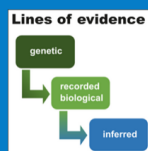
Please use the dropdown menu below for additional information.

- ▶ Subpopulations
- ▶ Evolutionarily Significant Units (ESU)
- ▶ Probable Evolutionarily Significant Unit (pESU)
- ▶ Gene Flow
- ▶ Potential IUCN Species assessment
- ▶ References

Scoring:

- 10 points or higher = distinct ESU
- Below 10 points:
 - a) 10 points cannot be reached or exceeded (even if all missing data were supportive of the tested scenario) = **no support for the ESU - Scenario.**
 - b) 10 points are **currently** not reached or exceeded, but could if missing data subsequently prove to be supportive of the tested scenario = **scenario indicates possible ESUs (pESUs)***

**However, if a candidate scenario currently only scores in recorded biological evidence-types (so non-genetic), the scenario should be regarded as 'data-deficient' rather than as a pESU scenario.*



Thank you , for participating in the **CGSG Subpopulation and ESU survey**.

The species you tested is the **Sclerophrys pantherina**

Your Survey Results for Phase 2:

1. Is there inherited variation in chromosome numbers, ploidy level or chromosome structure? (6 points)

1 ESU No data	2 ESUs No data	3 ESUs No data	
------------------	-------------------	-------------------	--

2. Is there evidence of long-term reproductive isolation? (6 points)

1 ESU Yes	2 ESUs Yes	3 ESUs No	
--------------	---------------	--------------	--

3. Is there evidence of local adaptation or adaptive divergence? (6 points)

1 ESU No data	2 ESUs No data	3 ESUs No data	
------------------	-------------------	-------------------	--

4. Is there evidence of inherited characteristic differences? (4 points)

1 ESU Yes	2 ESUs Yes	3 ESUs Yes	
--------------	---------------	---------------	--

5. Is there evidence supporting transmitted characteristic differences? (2 points)

1 ESU No data	2 ESUs No data	3 ESUs No data	
------------------	-------------------	-------------------	--

6. Is there traditional knowledge suggesting distinctiveness between units? (2 points)

1 ESU No data	2 ESUs No data	3 ESUs No data	
------------------	-------------------	-------------------	--

Total score for the scenario: **1 ESU** is 0

Total score for the scenario: **2 ESUs** is 10

Total score for the scenario: **3 ESUs** is 4

Total score for the scenario: is 0

Scenario scoring:

- Scenarios scoring 10 points or higher = distinct ESUs
- Scenarios scoring below 10 points:
 - **without** potential to reach/exceed 10 points = no support for tested scenario
 - **with** potential to reach/exceed 10 points = possible ESU (pESU) status

You'll be receiving the results in your inbox soon!

You've reached the end of the survey. We trust that the information will provide insight into the management of your species.

References

- Measey, G. J., & Tolley, K. A. (2011). Investigating the cause of the disjunct distribution of *Amietophrynus pantherinus*, the Endangered South African western leopard toad. *Conservation Genetics*, 12(1), 61–70. <https://doi.org/10.1007/s10592-009-9989-7>
- Measey, J., Tarrant, J., Rebelo, A., Turner, A., Du Preez, L., Mokhatla, M., & Conradie, W. (2019). Has strategic planning made a difference to amphibian conservation research in South Africa? *Bothalia*, 49(1). <https://doi.org/10.4102/abc.v49i1.2428>
- Rebelo, A. G., Holmes, P. M., Dorse, C., & Wood, J. (2011). Impacts of urbanization in a biodiversity hotspot: Conservation challenges in Metropolitan Cape Town. *South African Journal of Botany*, 77(1), 20–35. <https://doi.org/10.1016/j.sajb.2010.04.006>
- Stephens, K., Tolley, K. A., & da Silva, J. M. (2022). A decade of genetic monitoring reveals increased inbreeding for the Endangered western leopard toad, *Sclerophrys pantherina*. *Conservation Genetics*, 23(5), 903–918. <https://doi.org/10.1007/s10592-022-01463-5>